Bonding & properties

Bond particles

1 Three strong bonds

One shared pair → one bond

Ionic: attraction between opposite ions after electron transfer. Covalent: shared electron pairs attract both nuclei.

Metallic: positive ions attract delocalised electrons. Identify the charged particles in explanations.

2 Structure decides properties

Ionic lattices: high melting; conduct only molten or dissolved, when ions move.

Small molecules: low melting/boiling; weak forces between molecules. No mobile charged particles.

Giant covalent: many strong bonds → high melting.

Explain behaviour

3 States & models

Solid: particles vibrate in fixed positions. Liquid: close particles move past each other. Gas: particles far apart, moving randomly.

At a change of state, energy changes attractions/spacing, not particle identity. Model drawings simplify size, forces and movement.

4 Metals & polymers

Metals conduct through mobile delocalised electrons; layers can slide. Different-sized alloy atoms hinder sliding.

Polymers: long molecules, strong bonds along each chain, forces between chains; most are solid at room temperature.

Choose materials

5 Carbon structures

Diamond: four bonds per carbon, rigid and hard, no free electrons. Graphite: three bonds; weak forces between layers let them slide.

Graphene: one strong, conducting layer. Fullerenes: hollow cages; nanotubes: strong conducting cylinders.

6 Nanoparticles

1–100 nm: large surface area: volume ratio. Small amounts can be effective catalysts or coatings.

Uses include medicine, electronics, sunscreen and antibacterial surfaces. Properties differ from bulk material; assess exposure and uncertain risks.